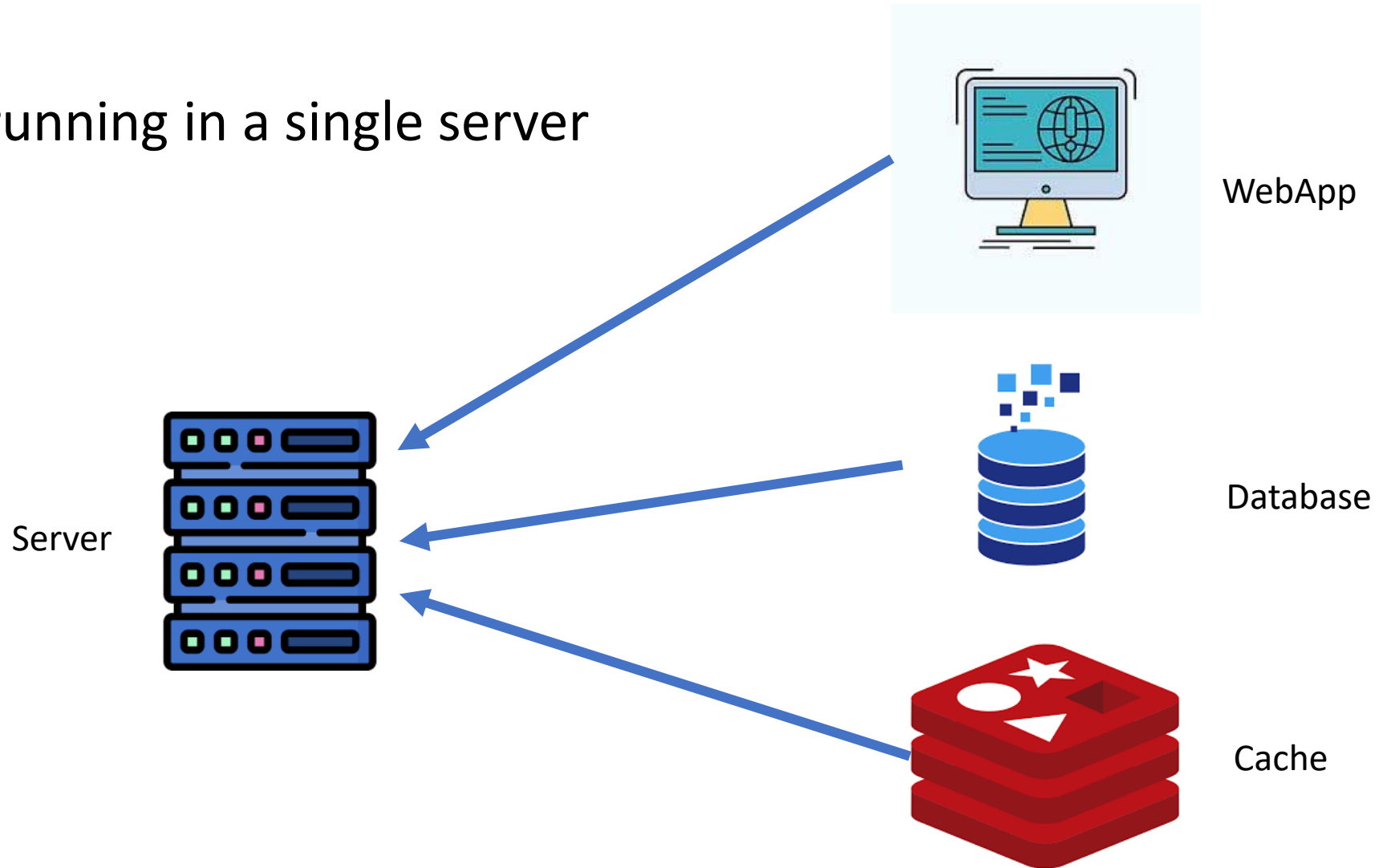


Evolution from Monolithic to N-Tier Architecture

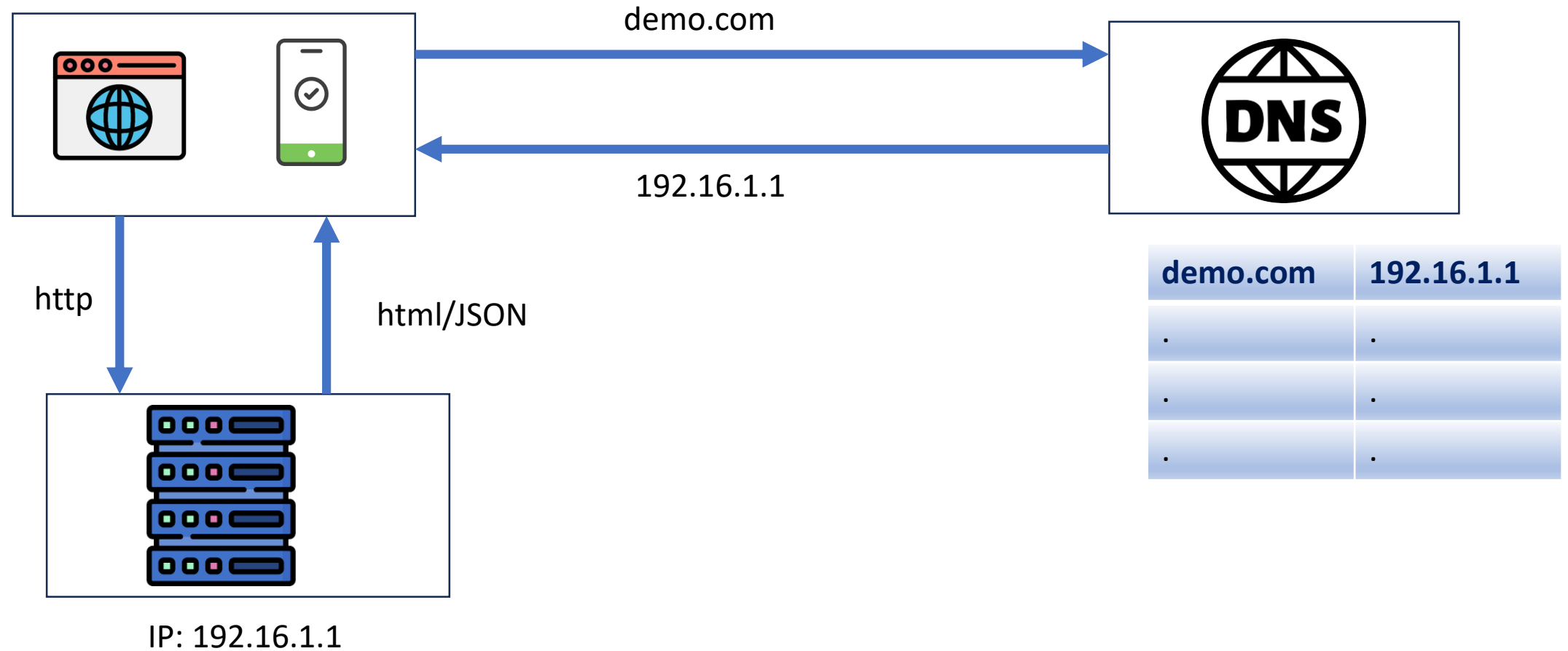
Prepared By: Dr. D. P. Singh

Design a Single Server Setup

- Everything is running in a single server



Handling of User Request



Traffic Sources

Web Application:

- Server handles data storage, business logic, presentation using HTML, CSS and JavaScript

Mobile Application:

- request data from server using API calls. Data is generally represented in JSON format

JSON

- JSON stands for **JavaScript Object Notation**.
- JSON is a **plain text format** for storing and transporting data.
- JSON is similar to the syntax for creating JavaScript objects.
- JSON is used to **send, receive and store data**.

JSON: Example

'{"name": "Ramesh", "age": 30, "car": null}'

The example above defines an object with 3 properties:

- "name"
- "age"
- "car"

Each property has a value:

- "Ramesh"
- 30
- null

Why JSON

- JSON makes it **easy to send and store data** between computers
- JSON is text only and **language independent**
- The syntax is **derived from JavaScript** object syntax, but JSON is text only.
- Code for reading and generating JSON data can be written in **any programming language**.

JSON and JavaScript

- The JSON format is syntactically identical to the code for creating JavaScript objects. Because of this, a **JavaScript program can easily convert** JSON data into native JavaScript objects.
- JavaScript has a built in function for converting JSON strings into JavaScript objects:

JSON.parse()

- JavaScript also has a built in function for converting an object into a JSON string:

JSON.stringify()

- You can receive pure text from a server and use it as a JavaScript object.
- You can send a JavaScript object to a server in pure text format.

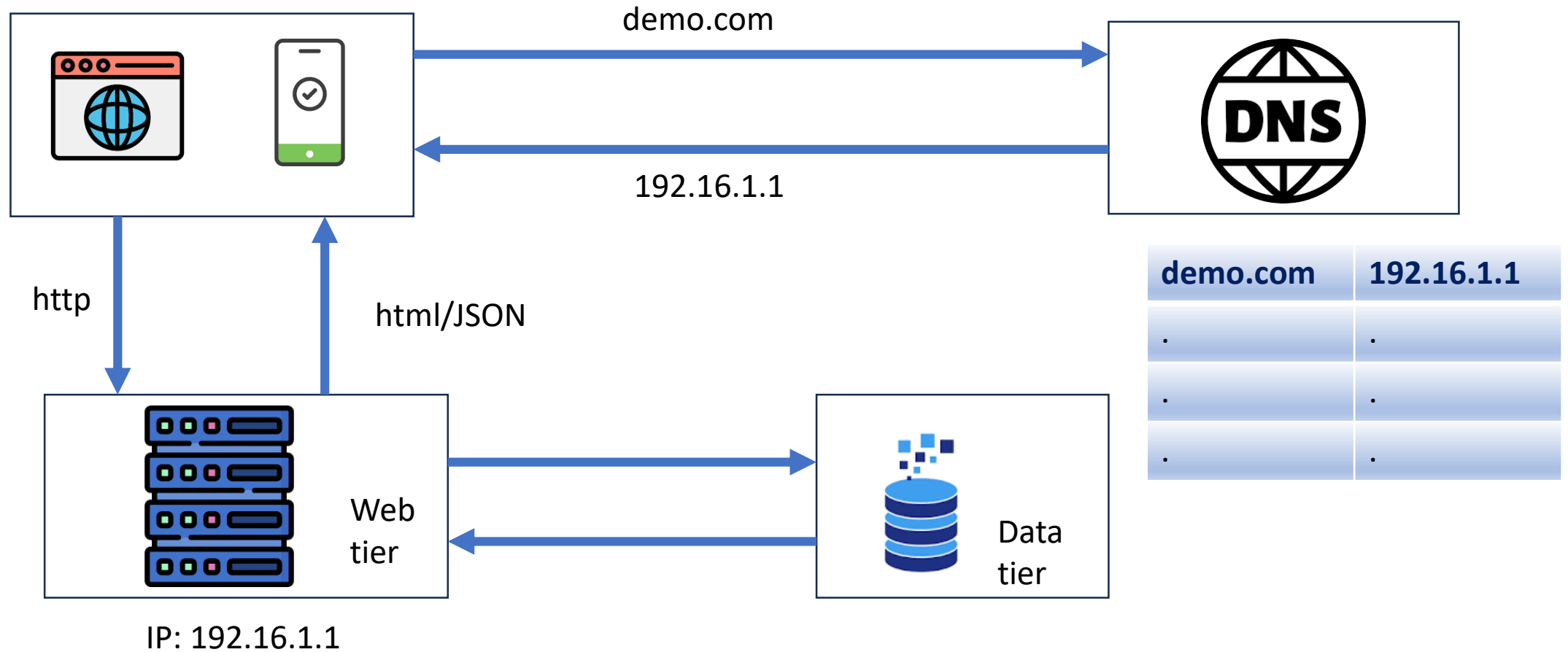
Single Server Setup short Falls

- Identify the short falls as user demand increases
- Take note:
 - Start from the single server system
 - Understand how requests flow through your system
 - **Traffic sources:** web and mobile app and **how they interact** with server

Strategies to optimize the setup

- When on. of users increases:
 - Divide the setup into two tier
 - Web tier
 - Data tier

Optimized setup



Thank You